Experiment: 8-bit Addition, Subtraction and Division Using 8085 Microprocessor

1. Aim

To write and execute 8085 assembly language programs to perform 8-bit addition, subtraction, and division of two numbers and store the results in specified memory locations.

2. Objectives

1. To understand 8-bit arithmetic operations using the 8085 microprocessor.
2. To apply appropriate arithmetic and data transfer instructions.
3. To store and verify the results using memory addresses.

3. Algorithm

A. Addition of Two 8-bit Numbers

1. Start the program.
2. Load the first 8-bit number from memory location `2050H`.
3. Store it in register B.
4. Load the second number from `2051H`.
5. Add register B to the accumulator.
6. Store the result at `2052H`.
7. Stop the program.

B. Subtraction of Two 8-bit Numbers

1. Start the program.
2. Load the first number from `2050H` into register B.
3. Load the second number from `2051H` into register C.
4. Move the first number into the accumulator.
5. Subtract register C from the accumulator.
6. Store the result at `2052H`.
7. Stop the program.

C. Multiplication of Two 8-bit Numbers

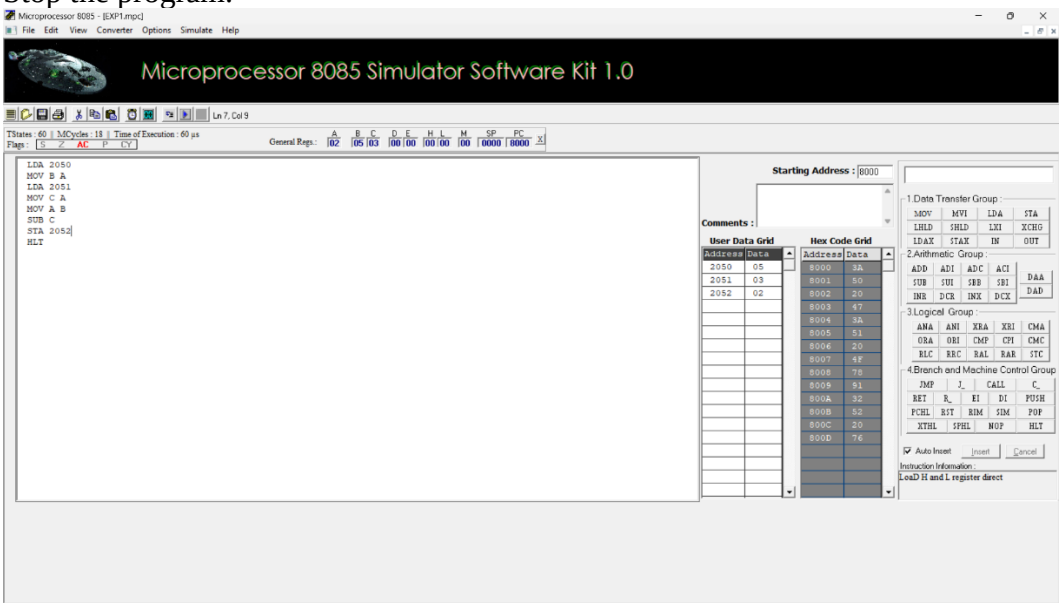
Algorithm

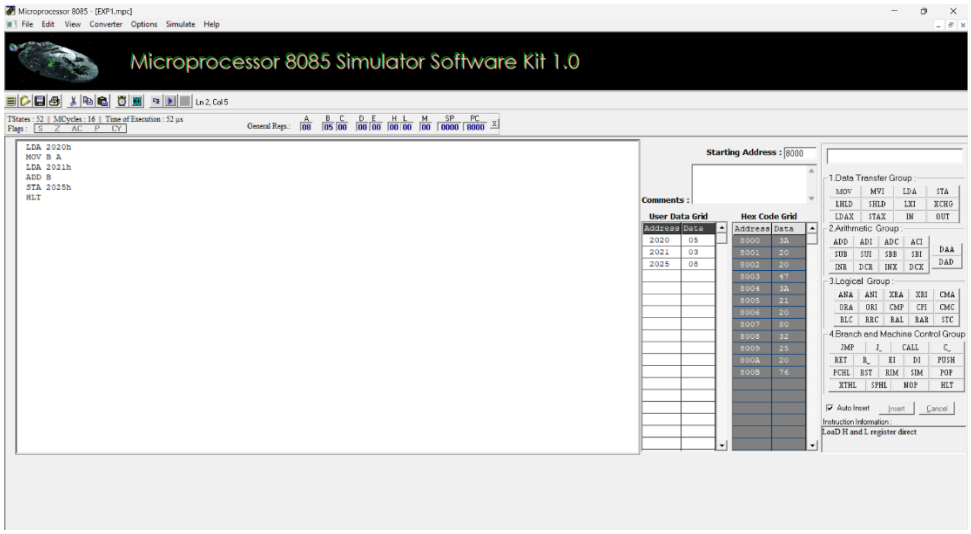
1. Start the program.
2. Load the first 8-bit number from memory location `2050H` into register B.
3. Load the second 8-bit number from memory location `2051H` into register C.
4. Initialize the accumulator to zero.
5. Repeatedly add the first number to the accumulator until the count reaches the second number.

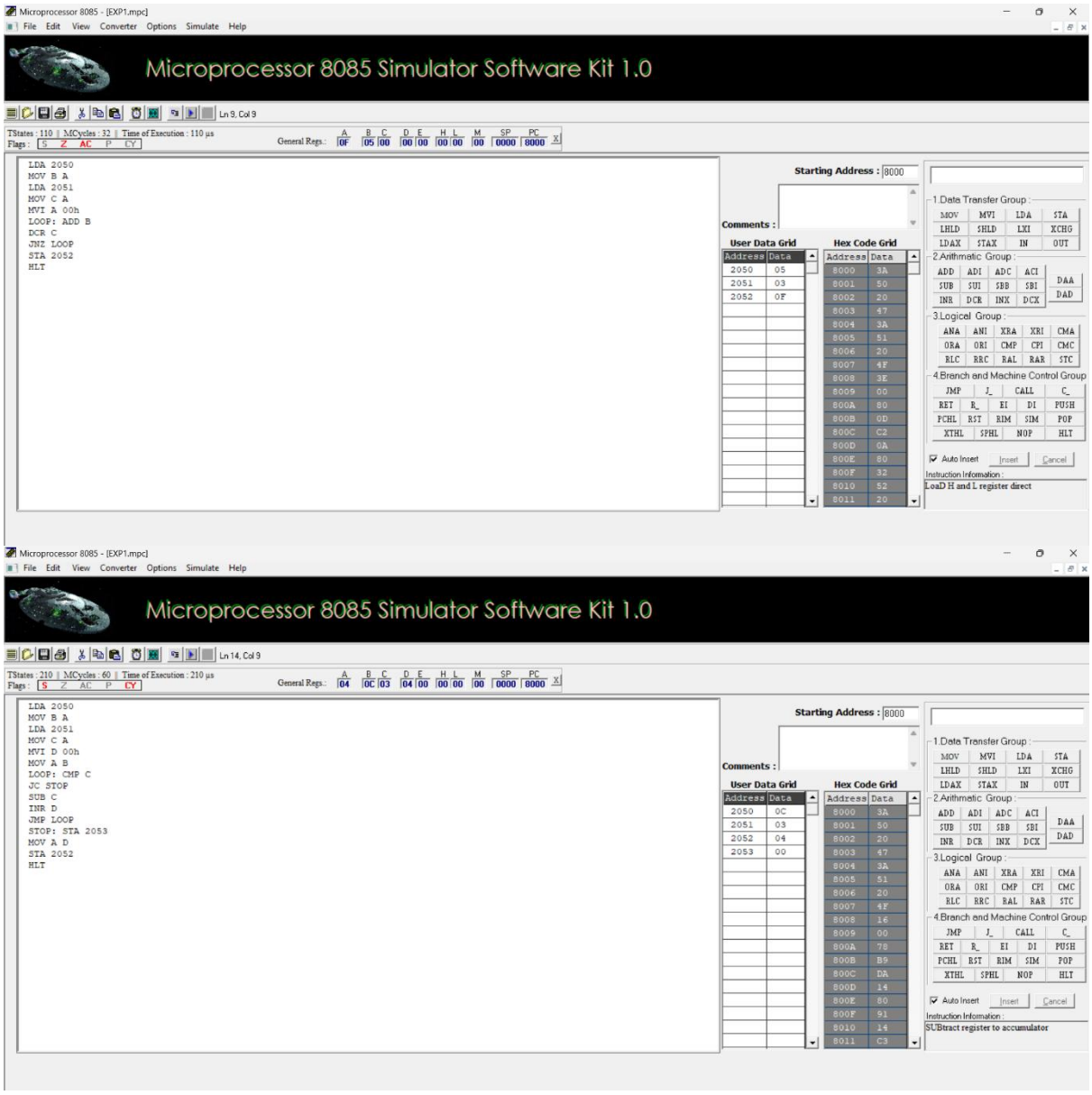
6. Store the 8-bit result at `2052H` and the carry (if any) at `2053H`.
7. Stop the program.

D. Division of Two 8-bit Numbers

1. Start the program.
2. Load the dividend from `2050H`.
3. Load the divisor from `2051H`.
4. Initialize the quotient to zero.
5. Repeatedly subtract the divisor from the dividend until the remainder is smaller than the divisor.
6. Store the quotient at `2052H` and the remainder at `2053H`.
7. Stop the program.







4. Result

The 8-bit addition, subtraction, and division operations were implemented using 8085 assembly language instructions. The results were stored in the specified memory locations and verified using the 8085 simulator.